C. All-in-one Gun

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: binary search, greedy, math | https://codeforces.com/problemset/problem/2192/C

You are developing a new shooter game, but since there are a lot of shooter games out there, you decide to have something unique in your game.

You have an all-in-one gun that shoots bullets in a fixed order. There are n bullets in the magazine, the i-th of which deals a_i damage. The enemy starts with h health and dies when its health becomes ≤ 0.

The gun shoots one bullet per second. After firing all n bullets, it must reload, which takes k seconds. Reloading always restores the **same sequence** of bullets $[a_1, a_2, \ldots, a_n]$. You cannot reload early; you must empty the magazine first. At the start, the magazine is already full.

Before the fight begins, you may perform **at most one** swap: pick any indices $1 \leq i$ lt $j \leq n$ and exchange a_i with a_j.

Your task is to find the minimum number of seconds needed to kill the enemy, taking into account this optional single swap.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each testcase contains three integer n, h and k ($2 \leq n \leq 2 \cdot 10^5$, $1 \leq h, k \leq 10^9$) — the size of magazine, health of your enemy and time required to reload the magazine.

The second line of each testcase contains n integers $a_1, a_2, \ldots, a_n$ ($1 \leq a_i \leq 10^9$).

It is guaranteed that the sum of n does not exceed $2 \cdot 10^5$ over all test cases.

Output

For each testcase, output a single integer denoting the minimum time required to kill the enemy.

Example

Input

```
6
5 10 1
4 2 3 5 3
5 10 1
4 2 3 7 3
3 10 2
1 2 3
2 5 3
2 1
3 18 5
1 2 3
4 10 10
1 1 2 2
```

Output
3 2 7 6 19 17

Note

In the first test case, you swap the bullets present at index 2 and 5. This makes array *a* as 4, 3, 3, 5, 2.

After 3 seconds, the health of your enemy will be 10 - 4 - 3 - 3 = 0, hence the enemy dies in 3 seconds. It can be shown that achieving time to kill less than 3 is not possible.

In the third test case, you swap bullets present at index 1 and 3. This makes array *a* as 3, 2, 1.

In 7 seconds, you shoot the entire first magazine (3 seconds) + reload a new magazine (2 seconds) + shoot the first and the second bullet from the new magazine (2 seconds).

The health of the enemy will be 10 - 3 - 2 - 1 - 3 - 2 = -1, hence the enemy dies in 7 seconds. It can be shown that achieving time to kill less than 7 is not possible.